

3–6 PERIODIC MAINTENANCE

3–6–1 When to Perform

Perform Periodic Maintenance (PM)

- at time of installation.
- each sixth month after installation in a mobile site.
- each twelfth month after installation in a fixed site.

3–6–2 How to Perform

All periodic maintenance can be accomplished through outer door and load panel. However, time required for completion may be reduced if at least one side or rear panel is also removed.

Perform a complete cleaning and inspection as described in Section 3–6–4, PM Procedure. Three hours are required to perform this procedure.

3–6–3 Tools Required

Common electrician tools are sufficient to complete this procedure, including torque wrench with allen sockets.

A recording torque wrench is needed if left side panel cannot be removed.

A vacuum cleaner with hose may be required.

3–6–4 PM Procedure



LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points – must measure 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.

3–6–4 PM Procedure (continued)

6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove and keep two nuts securing articulated arm to top of dead front.
8. Close dead front.
9. Remove and keep 10 screws securing dead front to load panel. Remove dead front covering load panel.
10. Check all circuit breaker connections for tightness. Torque to 30 lb in (34 Nm).
11. Inspect for dust. Vacuum any dust accumulation.
12. Put dead front on load panel and secure with 10 screws kept in Step 9.

Note

Verify that dead front will fit on the two guides at right when load panel door is closed.

13. Open load panel door. Install two nuts securing articulated arm to top of dead front.
14. Lock articulated arm.
15. Inspect circuit boards located on inner surface of load panel circuit breaker compartment and on regulation panel. See Illustrations 3–23 and 3–24.

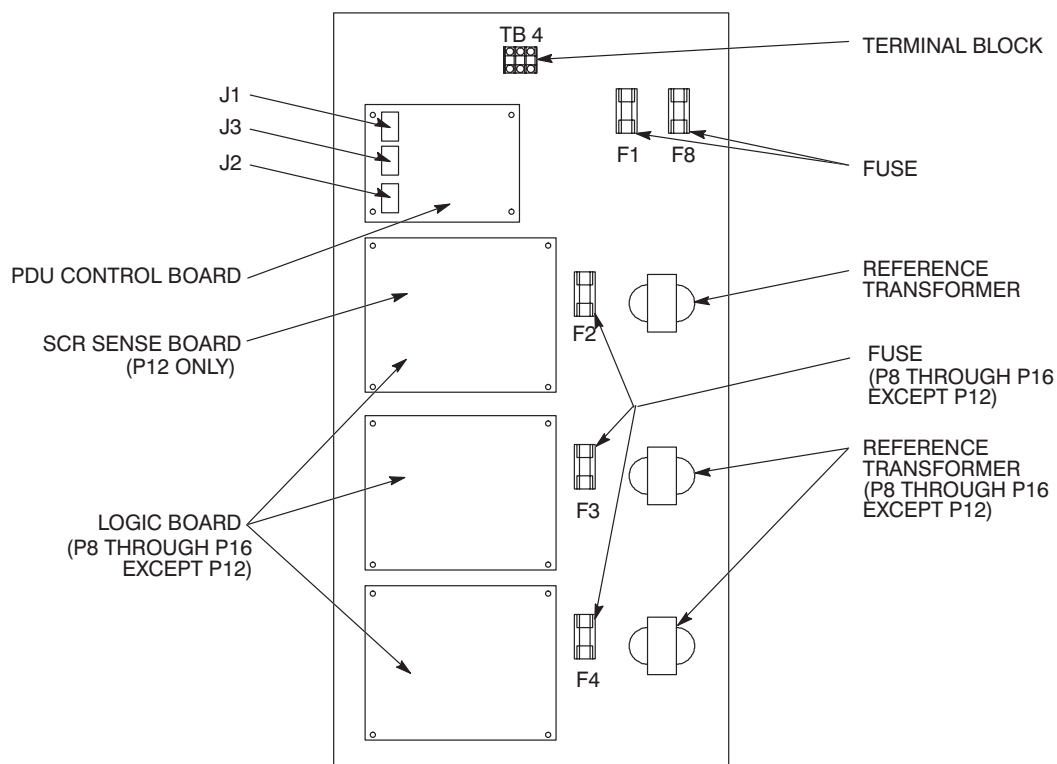
**LOAD CENTER BRACKET ASSEMBLY (A14)**

ILLUSTRATION 3–23

3–6–4 PM Procedure (continued)

- a. Inspect for evidence of overheating. Inspect for cracked, burned or distorted resistors, capacitors, and charred insulation. Replace any overheated parts.
 - b. Inspect for evidence of physical damage such as broken or cracked capacitors and resistors, broken solder joints and discontinuity in circuit foil paths.
 - c. Inspect circuit board connectors for corrosion.
 - d. Replace board if damaged. Refer to Section 3–4–1, Service a Circuit Board.
16. Inspect and torque all input power connections on SCRs, circuit breakers, switches, transformers, and filters. Inspect for discoloration or any other sign of excessive heating. Torque power connections as follows:
- a. Torque input power connections (eight screws) to 30 lb in (34 Nm).
 - b. Torque output power connections to 30 lb in (34 Nm).
 - c. Torque six main circuit breaker connections to 25 lb in (113 Nm). See Illustration 3–25.
 - d. Torque nine power wires on bypass switch. See Illustration 3–25. Torque to 30 lb in (34 Nm)

Note

With left side panel in place, use a recording torque wrench.

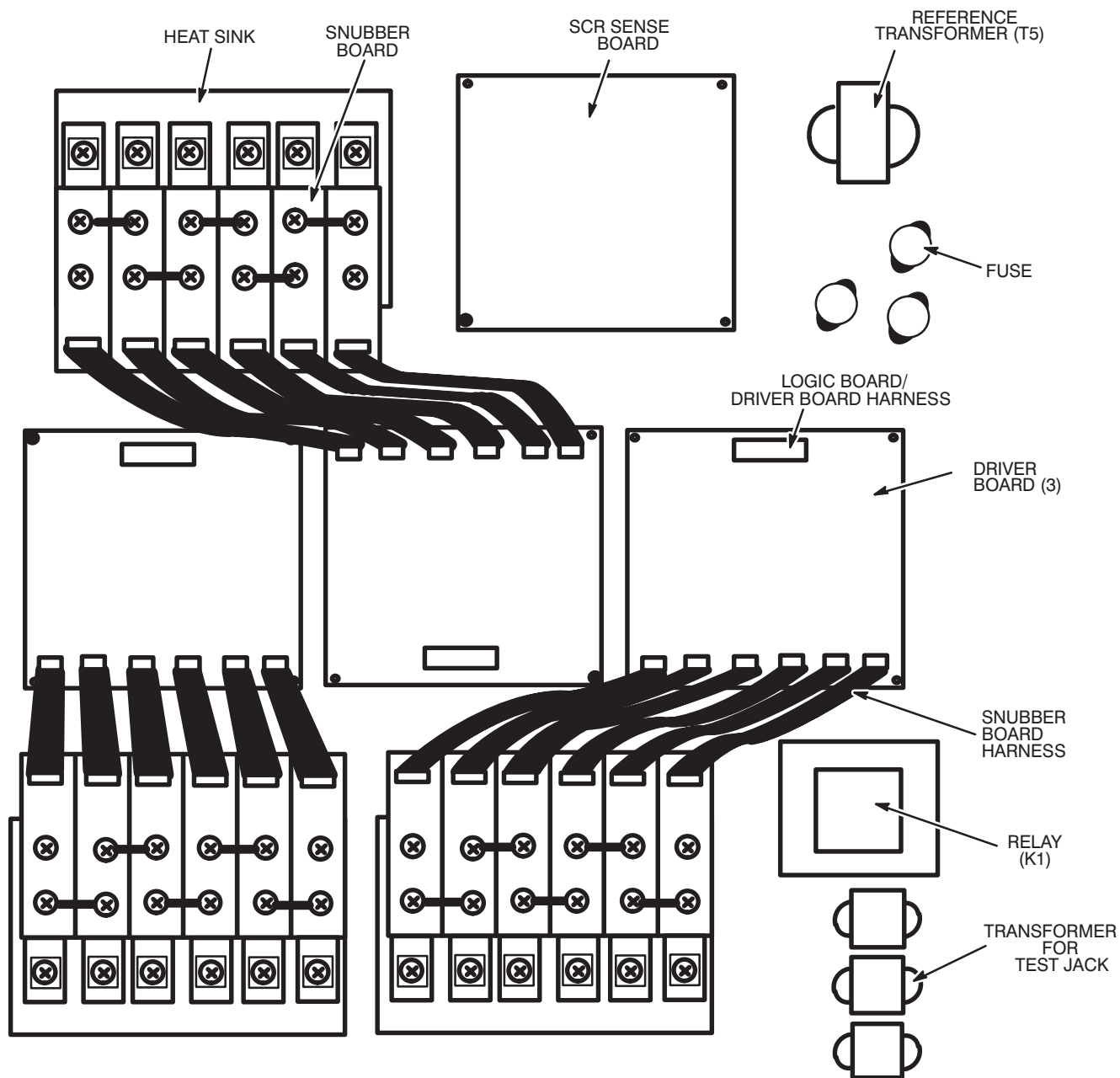
- e. Torque X_0 , X_1 , X_2 , X_3 and H_1 , H_2 , H_3 to 30 lb in (34 Nm). See Illustrations 3–14.
 - f. Models 46–317099 P8 through P16: Torque filter terminal block to 30 lb/in (34 NM) connections (A11A3, A11A4).
 - g. Models 46–317099 P8 through P16 except P12: Torque phillips screws on all 18 SCRs (54 screws) to 30 lb in (34 Nm). See Illustration 3–24.
17. Check connectors on circuit boards to insure proper seating. See Illustrations 3–23 and 3–24. This includes:
- a. Models 46–317099 P8 through P16 except P12: Connectors on both ends of 18 snubber board harnesses.
 - b. Models 46–317099 P8 through P16 except P12: Connectors on both ends of 6 logic board/driver board harnesses.
 - c. Nine connectors on SCR sense board.
 - d. Three connectors on PDU control board.
18. Check that relay A9K1 is properly seated. See Illustration 3–24 for location of relay.
19. Periodic maintenance for fan assembly (A7):
- a. Remove and keep four screws securing fan panel assembly to frame. Remove fan panel assembly.
 - b. Remove and keep fan mounting screws. Remove fan guard and fan.
 - c. Remove aluminum fan filters. Clean and install.

3-6-4 PM Procedure (continued)

Note

Mount fan so air flow is into Compact PDU.

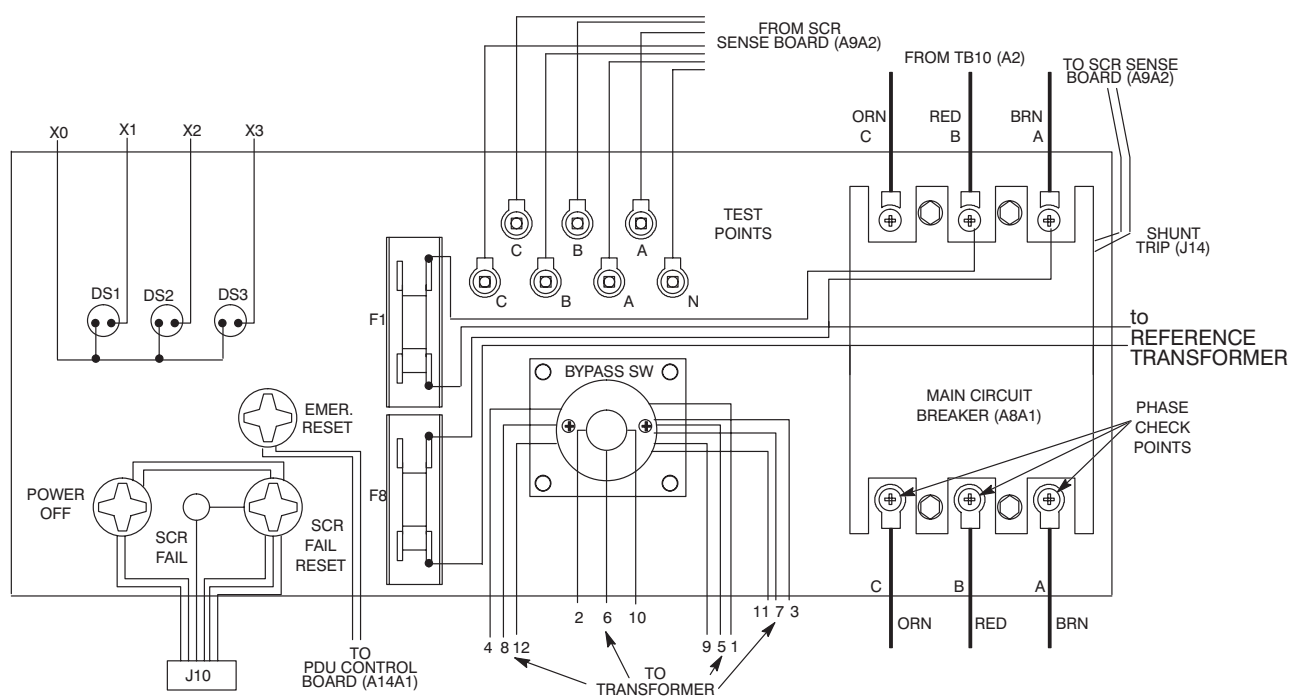
- d. Check that fan rotors are clean and rotate freely.
- e. Install fan guard and fan. Secure with fan mounting screws kept in Step 19-b.
- f. Install fan panel assembly and secure with four screws kept in Step 19-a.



REGULATION PANEL (MODELS 46-317099 P8 THROUGH P16 EXCEPT P12)

ILLUSTRATION 3-24

3-6-4 PM Procedure (continued)



FRONT PANEL (REAR VIEW)
ILLUSTRATION 3-25